Leakage-Aware Probabilistic Flood Forecasting for Bangladesh: A Spatiotemporal Evaluation of Machine Learning and Deep Learning Models

Abstract—Flood models can appear overly skillful when temporal dependence, station overlap, or forward-looking features leak information. We present a leakage-aware framework for probabilistic danger-level forecasting at 133 Bangladesh Flood Forecasting and Warning Centre (FFWC) stations over 24, 48, and 72 h. The 200,665 station-day dataset combines 3-hourly water levels with meteorological, geospatial, upstream-discharge, soil-wetness, and neighbouring-station information. Predictors obey an issue-time boundary, and evaluation uses chronological validation, rolling-origin refits, station-disjoint folds, and a frozen post-2022 test. Twenty-four baseline, classical-ML, and deep-learning configurations are compared through one scoring pathway. The selected per-horizon model (D16) achieves mean PR-AUC 0.9574 on 2022 validation and 0.7708 on the frozen test, versus 0.6675 for persistence, outperforming persistence on all 21 test cell–horizon points. The best deep configuration reaches 0.9356 on validation; a matched fusion rematch remains 0.0206 below XGBoost. The ECE is 0.003–0.009, and TreeSHAP assigns 64% of 24 h attribution to local hydrology. A protocol audit also determined and corrected an intermediary in-sample scoring defect without changing the chosen model or frozen-test result. The findings show that reliable flood AI evidence depends as much on leakage-aware evaluation and robust baselines as on model complexity.

***Index Terms*—Bangladesh, calibration, data leakage, deep learning, flood forecasting, gradient boosting, probabilistic prediction, SHAP, spatiotemporal validation.**

I. Introduction

Bangladesh is one of the most flood-exposed countries in the world, with riverine and coastal districts that depend on periodic warnings rather than on a single forecast of water level [1]. The Flood Forecasting and Warning Centre (FFWC) of the Bangladesh Water Development Board publishes station-specific warnings relative to a danger level: a fixed threshold above which a station is advised to be in flood [2]. Therefore, the concern is not the exact future stage in metres, but whether a station will cross that threshold in the next 24, 48, or 72 h. A probabilistic answer is more appropriate in this case because it allows warning authorities to trade missed events against false alarms.

Machine learning (ML) and deep learning (DL) are extensively used for short-term flood forecasting [3]–[5]. Studies in Bangladesh have examined rainfall–runoff processes, water-level regression, and climate scenarios [6]–[11]. More broadly, a range of methods, including gradient boosting, recurrent neural networks, temporal convolutions, attention mechanisms, and graph-based models, have been explored in flood forecasting research [3], [4], [12]. Although much of this work focuses on predicting continuous water level, evaluates models that use a single data split, or compares models with different input configurations, the results do not establish the skill needed for danger-level warning.

Three issues are central to warning-oriented evaluation. Hydro-meteorological autocorrelation makes random splits optimistic [13]; feature engineering and preprocessing can cause leakage [14], [15]; and accuracy or ROC-AUC can be misleading for drifting exceedance rates [16]. We therefore incorporate an issue-time information boundary, temporal and spatial refitting, consistent benchmarking, calibration, and event-oriented metrics.

We make three contributions. First, we formulate 24/48/72 h station-specific danger-level exceedance with leakage-controlled rolling-origin and station-disjoint evaluation, including a documented protocol correction. Second, we compare 24 configurations that include seven baselines, five classical models, and twelve DL configurations under common partitions, labels, seeds, metric code, and tuning budget. Third, we assess the selected model against persistence on trained, new-network, and unseen stations using calibration, cost-sensitive thresholds, event detection, and TreeSHAP. The objective is a decision-oriented benchmark rather than a claim that one architecture is universally superior.

II. Related Work and Research Gap

Data-driven flood forecasting now employs tree ensembles, recurrent and convolutional sequence models, transformers, and graph networks [12], [17]–[26]. Long short-term memory (LSTM)-based rainfall–runoff modelling has demonstrated strong sequence-learning capability [21], [38], while XGBoost and LightGBM remain competitive for engineered tabular predictors [17], [18]. Reviews progressively emphasise interpretability and reliable out-of-sample testing alongside predictive accuracy [39], [40].

Recent work in Bangladesh has shown effective signal for water-level and discharge prediction and has begun to examine spatial transferability [6]–[11]. However, the operational FFWC task is different from generic regression: the goal is a rare threshold crossing for specific stations, miss and false-alarm costs differ, and useful evidence must differentiate future-time from unseen-station generalisation.

Therefore, the research gap is methodological: leakage-aware, multi-horizon, station-level evidence that uses a persistence reference, segregates temporal and spatial generalisation, reports calibration and event detection, and explains predictions hydrologically remains limited [13], [16], [29]–[32]. This gap is addressed by treating evaluation design as a first-class contribution and by reporting an internal protocol correction transparently.

III. Dataset and Forecast Formulation

A. Data Sources and Coverage

Primary data include the FFWC station master and 1,052,044 three-hourly water-level observations between January 2017 and October 2026. Although the dataset initially included 153 stations, 133 had both a usable danger level and at least one valid label and were retained; the year 2018 was excluded because of missing data (Fig. 1a). Predictor variables were compiled from different sources, including NASA POWER/MERRA-2 meteorological data [37], SRTM

topographic data [36], OpenStreetMap and geoBoundaries attributes [41], [42], GloFAS discharge data [35], and POWER soil moisture. ERA5-Land [34] was tested but not retained.

Three-hourly observations were aggregated into daily summaries using a complete calendar of station-days; these features included daily maximum, mean, and minimum water levels, the last observed reading, and the number of available readings per day. After quality control, the dataset contains 200,665 station-days across 133 stations. The number of stations in the network progresses from 85 in 2017 to 133 by the end of the record. This increase is reflected in the validation setup (Table I).

TABLE I
Data Partitions and Danger-Level Base Rates

Partition	Rows	Stations	24 h	48 h	72 h
Training	77,571	105	9.60	10.72	11.73
Validation	21,920	105	5.08	5.96	6.64
Frozen test	101,174	133	1.41	1.77	2.12

Base rates are percentages of labelled station-days. The test period is 2023-01-27 to 2026-10-05; 2018 is absent from the FFWC archive.

Note: The training and validation rows are full partition counts (105 stations). Of the 26 station-holdout stations, 21 have pre-2023 rows; excluding those 21 leaves 84 stations and 61,991 training / 17,538 validation rows. The frozen test (101,174 rows, 133 stations) is unchanged.

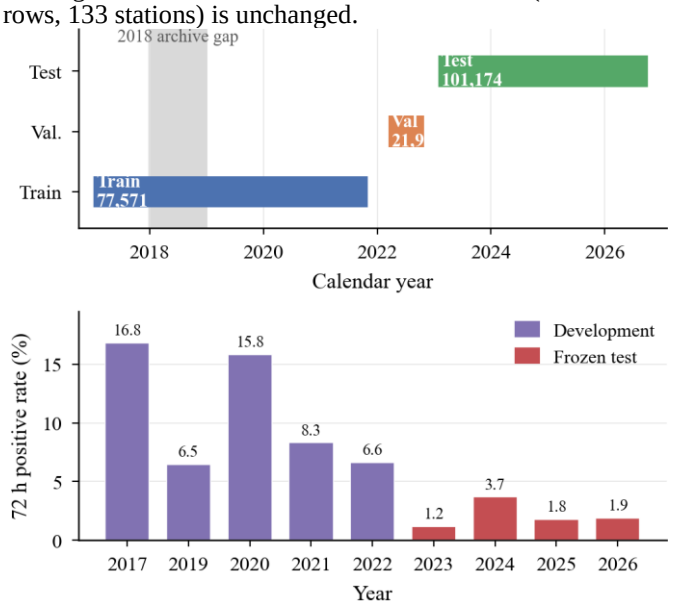


Fig. 1. Temporal coverage and label prevalence. (a) Training, validation, and frozen-test spans; the 2018 archive gap is preserved as missing. (b) Annual 72 h positive rate, showing the 14-fold decline from 2017 to 2023 and the higher prevalence of the development years. Cell-level, not pooled, evaluation follows from this drift.

B. Forecast Formulation and Labels

Let s index a station, t an issue day, and $h \in \{24,48,72\}$ the lead time in hours. The model estimates the probability that station s reaches its danger level D_s within the next h hours, using only information available at the end of day t:

$$P_{s,t,h} = \Pr(Y_{s,t,h} = 1 \mid I_{s,t}) \quad (1)$$

where $I_{s,t}$ is the issue-time information set. The target label indicates whether any observed 3-hourly water level reading reaches or outpaces the danger level within the forecast horizon for each station:

$$Y_{s,t,h} = 1\{\max_{u \in (t,t+h]} W_{s,u} \geq D_s\} \quad (2)$$

A negative label is defined only when the whole forecast window has ample observation coverage and no recorded water level reading reaches the station's danger level. Positives are retained when any observed exceedance occurs. Valid labels number 200,559, 197,748, and 195,174 at 24, 48, and 72 h, with positive rates of 4.98%, 5.68%, and 6.32%. Strong temporal drift—the 72 h rate falls from 16.81% in 2017 to 1.19% in 2023 (Fig. 1b)—and a 5.4× monsoon/non-monsoon contrast motivate cell- and season-specific reporting.

C. Spatiotemporal Partitions

Random splitting was avoided to preserve the temporal structure of the data. The chronological split assigned observations from 2017–2021 to training (77,571 rows), 2022 to validation (21,920 rows), and 2023–2026 to the frozen test set (101,174 rows). Natural dry-season gaps of 88–131 days separate the periods; a seven-day embargo is retained as a safety rule.

Spatial generalisation was examined using five station-disjoint folds. Fold 0 is the frozen holdout of 26 stations that remained unseen, whereas folds 1–4 served model development. Test stations are also classified as trained, new-network, or unseen. Five protocols are used: P1 primary 2022 validation; P2 rolling-origin 2020–2022 refits; P3 four-fold station-disjoint refits; P4 frozen unseen-station testing; and P5 monsoon/non-monsoon stratification. These protocols defined seven development cells: val_2020, val_2021, val_2022, and spatial folds 1–4.

IV. Methodology

A. Features and Leakage Control

FULL contains 62 predictors spanning weather, rainfall history, geography, local hydrology, and the station network. With the addition of soil moisture, GloFAS flow, water-level persistence, and antecedent precipitation features, the controlled dataset FULL_V2 holds 110 features. Hydrological features include relative water levels, distance to danger, recent maxima, rise rates, exceedance counts, and antecedent rainfall.

All features used on issue day t capture only information available at the end of day t. Any IDs, dates, labels, origin information, and future observations are excluded; lagged and rolling features are always backward-looking; preprocessing is fitted within each training fold; and sequence-based features cannot overlap across station or split boundaries. Neighbour features are created on the operational station network prior to assigning folds. As a result, rows of the target variable are disjoint between stations, but the covariates are not completely disjoint. This is noted as a shortcoming.

B. Models and Tuning

We compare 24 model configurations: seven baseline models (including D7 hydrological persistence), five classical models—random forest [19], gradient boosting [20], XGBoost [17], LightGBM [18], and XGBoost with feature selection—and twelve deep learning configurations (LSTM [21], gated recurrent unit (GRU) [22], temporal convolutional network (TCN) [23], fusion models, graph convolutional network (GCN) [25], graph attention network (GAT) [26], graph neural network (GNN)–LSTM, and Transformer [24]). Models were trained separately for each horizon. Classical models were trained using tabular features, while sequential models use a 12-variable raw sequence and fusion models combine both.

Hyperparameters are fixed with common splits, labels, seeds, and metric code. D16 shows the highest horizon-mean PR-AUC on val_2022 while achieving higher PR-AUC than D7 in each development cell with ECE ≤ 0.02; it was selected using LightGBM (24/48 h) and XGBoost (72 h) on FULL_V2. A supplementary matched fusion rematch had the same

engineered vector, five anchors, three seeds, and station bootstrap; because only the fusion family completed this rig, it is confirmatory rather than a full DL rerun.

C. Metrics and Decision Rules

The area under the precision–recall curve (PR-AUC), measured using average precision (AP), was selected as the primary evaluation metric for each forecast horizon, evaluation cell, and season [16], and ROC-AUC was reported as the secondary metric. Average precision was calculated as

$$\mathrm{AP} = \Sigma_n (R_n - R_{n-1}) P_n \quad (3)$$

where P_n and R_n stand for precision and recall at the n-th threshold. Operational performance was assessed using recall, precision, the number of missed flood rows, and the false-alarm rate at a cost-sensitive threshold. Calibration was evaluated using the Brier score [31] and expected calibration error with ten equal-width bins:

$$\mathrm{ECE} = \Sigma_{b=1}^{B} (n_b/N) \, |\bar{y}_b - \bar{p}_b| \quad (4)$$

The operating threshold is chosen to minimize the row cost

$$C(\tau) = C_{FN} \, \mathrm{FN}(\tau) + C_{FP} \, \mathrm{FP}(\tau) \quad (5)$$

with C_{FN}:C_{FP} = 10:1 for the main experiment and 5:1 and 20:1 for assessing sensitivities. These thresholds were selected using development data and kept unchanged before evaluation on the test set. Four research-defined risk categories were established as low, moderate, high, and very high. These categories are only for research analysis and do not represent the official FFWC warning levels. For event-level evaluation, successive positive days at each station were grouped into events, with gaps of one day bridged. In the frozen-threshold test evaluation, an event was considered detected if an anomaly occurred within the interval [start − h, end]. The development evaluation used a stricter event definition: each event had to be preceded by a warning, events that could not be verified were excluded, and false-alarm episodes were counted only when they fell outside the event and lead-time window.

Model explanations were achieved using exact TreeSHAP values for each frozen boosting model [27], [28]. SHAP values are aggregated to the nine feature groups, and cross-model agreement is measured by Spearman rank correlation against the Phase-33 XGBoost incumbent. Permutation importance is computed as a second opinion, but it is known to be unstable under collinearity; disagreements are reported rather than hidden.

D. Reproducibility and Test Discipline

All runs use seed 20261008; uncertainty uses station bootstrap [33], and scripts and configurations are archived with a SHA-256 manifest. The programme-level frozen test was opened six times: five predictive reads (one of them invalid and retracted) and one label-statistics read. D16's reported MI-6 read was pre-registered and reproduced bit-for-bit by MI-9. Thus the test is frozen for D16 but not pristine for the programme, and this history is disclosed explicitly.

V. Results

A. Classical ML versus Deep Learning

Table II and Fig. 2 report validation PR-AUC on P1 val_2022 for representative models; the full 24-configuration table is archived with the analysis code. The best classical model is XGBoost, with a horizon mean of 0.9546 (0.9681, 0.9549, and 0.9408 at 24, 48, and 72 h). LightGBM is statistically indistinguishable at 0.9534, and the top four classical models lie within 0.0023 of one another. The fusion model stands highest among all deep learning models at 0.9356 (0.9444, 0.9385, and 0.9238), which is 0.0190 below XGBoost. All five tree-based classical models outperform every deep learning model at all horizons. However, not all deep learning models are equal: the transformer and GCN models rank second and third among the DL configurations, below fusion and above the remaining models.

TABLE II
P1 Validation PR-AUC on val_2022 (all season)

Model	24 h	48 h	72 h	Mean
D7 persistence	0.9203	0.8954	0.8683	0.8947
Climatology	0.0509	0.0594	0.0658	0.0587
Logistic regression	0.9401	0.9241	0.9053	0.9232
Gradient boosting	0.9618	0.9457	0.9338	0.9471
Random forest	0.9665	0.9516	0.9389	0.9523
XGBoost-selected	0.9663	0.9533	0.9422	0.9539
LightGBM	0.9682	0.9538	0.9384	0.9534
XGBoost	0.9681	0.9549	0.9408	0.9546
LSTM-3	0.9421	0.9253	0.9129	0.9268
GRU	0.9359	0.9263	0.9086	0.9236
TCN	0.9405	0.9224	0.9008	0.9212
Fusion	0.9444	0.9385	0.9238	0.9356
GCN	0.9421	0.9353	0.9141	0.9305
GAT	0.9408	0.9315	0.9136	0.9286
GNN–LSTM	0.9424	0.9264	0.9153	0.9280
Transformer	0.9437	0.9330	0.9183	0.9317

Representative subset of the 24 configurations. D16 is the selected per-horizon ensemble (0.9708/0.9565/0.9450; mean 0.9574), not a single row in this table.

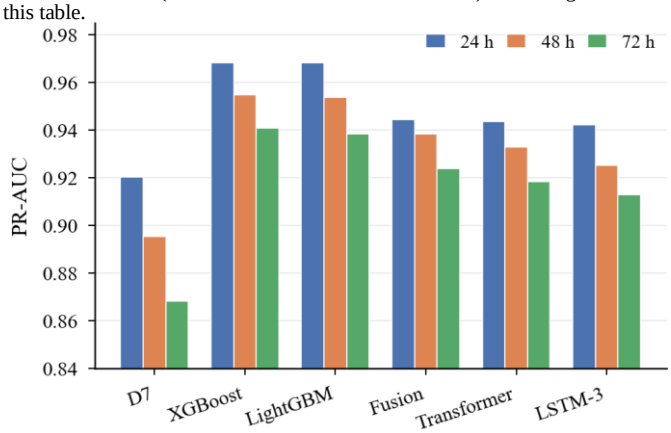


Fig. 2. Validation PR-AUC by horizon for representative models. Gradient-boosted trees occupy the top of every horizon; the fusion model is the strongest DL configuration but remains below the classical leaders. Values are from the P1 val_2022 cell.

The cell-level confidence intervals overlap, partly reflecting the limited validation coverage, with only 84 validation stations contributing data for a single year. However, paired bootstrap comparisons consistently demonstrated better PR-AUC performance for XGBoost than for the matched fusion model across all three forecast horizons. The gaps in PR-AUC (fusion minus XGBoost) are −0.0228, −0.0190, and −0.0165, respectively, and all corresponding 95% confidence intervals exclude zero. The mean difference is −0.0194. The fusion model shows good calibration, but its weaker ranking performance and higher false-alarm rate indicate that fusing the model inputs does not necessarily improve predictive performance.

Model complexity does not improve this task: the fusion model has 58,817 weights and is slower than XGBoost-selected, while its spatial-fold performance degrades more sharply. Engineered lag features therefore capture local persistence more efficiently than the tested DL representations.

B. Selected Model and Frozen Test

D16 achieves 0.9708, 0.9565, and 0.9450 on val_2022 (mean 0.9574), marginally above the best single classical model because it selects the best per-horizon learner rather than forcing one family. On the frozen test (Table III; Fig. 3), D16 achieves 0.8362, 0.7690, and 0.7071 (mean 0.7708), against D7 persistence at 0.7340, 0.6625, and 0.6062 (mean 0.6675). The performance of D16 is higher than persistence for each cell–horizon pair. The smallest improvement is recorded at unseen stations (72 h, +0.0424).

TABLE III
Frozen-Test PR-AUC: D16 versus D7 Persistence

Test cell	24 h	48 h	72 h	Mean Δ
2023	0.796 / 0.628	0.681 / 0.552	0.626 / 0.511	+0.137
2024	0.918 / 0.802	0.857 / 0.730	0.804 / 0.673	+0.125
2025	0.805 / 0.733	0.760 / 0.667	0.702 / 0.604	+0.088
2026	0.794 / 0.723	0.747 / 0.649	0.702 / 0.593	+0.092
Trained	0.869 / 0.747	0.809 / 0.676	0.761 / 0.618	+0.133
New network	0.843 / 0.736	0.787 / 0.693	0.713 / 0.646	+0.090
Unseen	0.830 / 0.770	0.742 / 0.669	0.641 / 0.599	+0.058
Mean	0.836 / 0.734	0.769 / 0.663	0.707 / 0.606	+0.103

Each cell shows D16 / D7. Mean Δ is the mean of the three per-horizon differences, not a difference of means. All 21 D16–D7 cell–horizon comparisons favour D16.

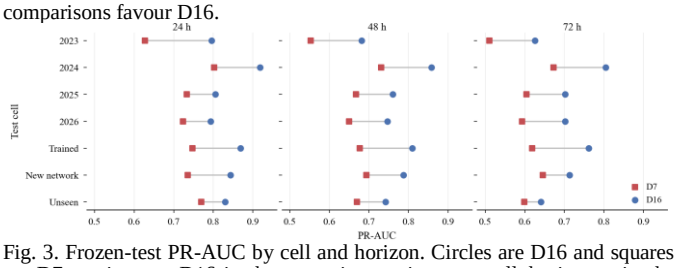


Fig. 3. Frozen-test PR-AUC by cell and horizon. Circles are D16 and squares are D7 persistence. D16 is above persistence in every cell–horizon pair; the smallest margin is unseen stations at 72 h (+0.0424).

Test scores are lower than validation because prevalence changes sharply: the 24 h base rate falls from 5.08% to 1.41%. D16 still exceeds persistence for all test cell–horizon pairs, but performs least well at 72 h and 2023. This variation highlights the importance of temporally separated evaluation.

D16 outperforms XGBoost in 6 of 7 test cells; XGBoost is better only in 2025, with a slightly higher 72 h mean (0.7092 vs. 0.7071). Gains over persistence are therefore consistent, although D16 does not beat every classical alternative.

The model is well calibrated, with a mean ECE of 0.0028, 0.0042, and 0.0085 at 24/48/72 h and all 21 cell–horizon pairs below 0.02. At the frozen 10:1 thresholds, pooled recall is 0.918, 0.903, and 0.856 and pooled precision is 0.549, 0.388, and 0.351. The best operating point will be selected by the warning service.

During validation, approximately 5–6% of rows fell in the very high risk category, with an actual flood rate near 90%; these are only research categories, not FFWC warning levels. Under the lenient frozen-test event criteria, the model detected 93.0%, 96.5%, and 96.9% of events at 24, 48, and 72 h, versus 86.4%, 92.7%, and 91.2% for D7.

C. Temporal and Spatial Generalisation

After the preregistered correction, D16 has a higher score than D7 on all 21 development cell–horizons (minimum +0.0156). The corrected average score is 0.9665 across the seven cells; however, this should be interpreted descriptively due to near saturation of four spatial folds and a much lower test base rate.

The three classical leaders beat D7 in all 63 all-season development comparisons, but not under every stress condition: extreme rainfall and a stricter one-day issue-time shift produce some losses. The appropriate claim is therefore robust all-season improvement, not universal dominance.

The ablation study shows that adding local hydrology to weather and geography raises mean PR-AUC by +0.4702 (range +0.389 to +0.539), whereas the station-network features contribute only +0.0009 and dropping four upstream columns changes the mean by −0.0001. Therefore, the model is mainly driven by local hydrological features and rainfall history. This is supported by the SHAP analysis and by the small increases from the GNN.

D. Protocol Audit and the P2/P3 Correction

The protocol audit demonstrated that the intermediate MI-8 probe scored six of the seven development cells using a model already fitted on those rows. Correct per-cell refitting reduces the seven-cell means for the 62-feature FULL, 110-feature FULL_V2, and 165-feature FULL_V3 models from 0.9834, 0.9868, and 0.9882 to 0.9641, 0.9659, and 0.9678, respectively (Table IV). The feature audit still passed, and val_2022 and the frozen test were untouched.

TABLE IV
P2/P3 Re-Scoring Correction

Quantity	Archived	Corrected	Change
FULL, 7-cell mean	0.9834	0.9641	−0.0194
FULL_V2, 7-cell mean	0.9868	0.9659	−0.0209
FULL_V3, 7-cell mean	0.9882	0.9678	−0.0204
FULL_V3 − FULL_V2	+0.00136	+0.00187	+0.00051
V3 wins, cell–horizon points	21/21	20/21	−1
D16 − D7 floor	+0.0225	+0.0156	−0.0069

The archived values scored six cells with a model fitted on those cells. Corrected values refit each cell. The champion and the frozen-test read are unchanged because val_2022 was always out of sample.

The corrected scores show that FULL_V3 exceeds FULL_V2 by +0.00187 and wins 20/21 cell–horizon points; the minimum gap between D16 and D7 is now +0.0156. This episode is important because leakage-aware evaluation must audit scoring masks as well as predictor construction.

E. Strict Event-Level Evaluation

The strict development analysis asks whether a warning is issued before an event begins, not merely somewhere in the event window. D16 has higher mean recall than D7 at all horizons, with 50.9%, 72.4%, and 88.1% at 24, 48, and 72 h, respectively; D7 had 45.2%, 64.9%, and 76.7%. D16 also produces fewer false-alarm episodes per station-year at every horizon (Table V), and its recall advantage grows with lead time. At 24 h, however, D16 loses strict recall on two development cells (val_2020 and spatial fold 1), and the mean margin is only 5.6 percentage points; it is not uniformly better at short lead times.

TABLE V
Strict Event-Level Performance on Development Cells

Horizon	D16 recall	D7 recall	D16 FA	D7 FA
24 h	50.9%	45.2%	0.58	1.56
48 h	72.4%	64.9%	0.98	1.97
72 h	88.1%	76.7%	2.30	2.44

Mean across all seven development cells. Recall requires a warning before event onset; FA is false alarms per station-year. No additional frozen-test read was performed.

The relaxed event definition results in higher recall by including any alert within the lead/event period. However, the stricter definition counts false-alarm episodes outside the event periods. Considering both metrics separately avoids any claim that one definition benefited from the other.

F. Hydrologically Interpretable Explanations

The local hydrology group accounts for 64.0% of mean absolute 24 h attribution using TreeSHAP in the frozen D16 boosters (Fig. 4). Geography, network, weather, GloFAS, and interactions contribute 8.4%, 6.1%, 4.7%, 4.6%, and 4.1%, respectively, whilst soil moisture, flow memory, and rainfall history contribute 4.1%, 2.4%, and 1.5%. The top individual features are current water level as a fraction of danger level, distance to danger, seven-day maximum water level, upstream water level, and 72 h rainfall. This is intuitive to a hydrologist: local stage dominates the analysis, followed by upstream and rainfall information providing context.

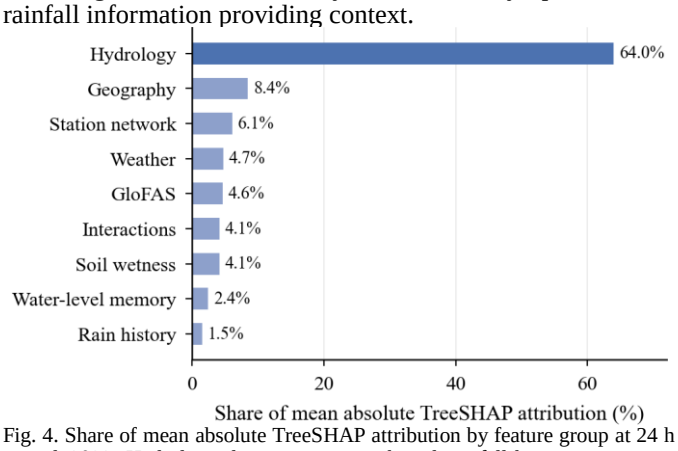


Fig. 4. Share of mean absolute TreeSHAP attribution by feature group at 24 h on val_2022. Hydrology dominates; network and rainfall-history groups are small once local stage is known.

Feature-group explanations are stable across models (Spearman ρ = 0.755–0.878 between D16 and the earlier XGBoost incumbent). Permutation importance showed some variation at the 24 h horizon, likely because of correlations among predictor variables. The interpretation therefore focuses on feature-group TreeSHAP results rather than individual feature rankings.

VI. Discussion and Limitations

A. Why Gradient Boosting Wins Here

These findings suggest that model performance depends on the characteristics of the forecasting task and the available input data. Predicting whether water levels exceed danger thresholds is a rare-event classification problem involving tabular, temporally correlated observations and engineered lag features. Gradient-boosted trees use the threshold nature and persistence effectively. Although the fusion rematch reduces representation differences, it still underperforms XGBoost by 0.0206. This result should not be interpreted as evidence that deep learning is unsuitable for flood forecasting in general. Rather, it suggests that the available inputs may not provide sufficient additional information to justify the greater model complexity in this particular case.

The ablation results support this interpretation: local hydrological features provide the largest contribution to most of the predictive skill, whereas network-based features provide relatively limited additional benefit once local water-level information is available. Together, these findings highlight the importance of reliable observations, consistent label definitions, and informative hydrological features when developing flood forecasting models. For this reason, these factors appear more consequential than simply increasing architectural complexity in this particular case.

B. What the Protocol Audit Changes

The issue in P2/P3 was identified as validation leakage rather than feature leakage. Refitting the model separately for each evaluation cell lowers the development scores by approximately 0.02. However, the champion model and the frozen test results were unchanged, since val_2022 remained out of sample throughout the evaluation. This highlights the need for automated leakage checks to assess evaluation masks, data-splitting procedures, and predictor features.

The test ledger also stated that the programme entered the frozen testing period before D16 had been finalised. Though the preregistered evaluation of D16 followed the specified procedure, the test history was not entirely clean. This experience underscores the importance of transparent reporting in benchmark studies, including the timing of each test-set opening, the model version evaluated, and whether the resulting scores influenced model selection.

C. Operational Interpretation

The results suggest that D16 is better used to supplement rather than replace existing models for operational flood forecasting. It is better than persistence at 24 h, though precision at the 10:1 threshold is still 0.55. At 72 h it is less competitive, with the earlier XGBoost model slightly higher. It does not use any forecast rainfall or numerical weather prediction (NWP) data, so its predictions are based on observed water level, upstream propagation, and recent rainfall. For this reason, the most apparent path to longer-lead skill is forecast rainfall.

Cost-sensitive thresholds offer an explicit mechanism to audit the trade-off between flood events and false alarms. The 5:1–20:1 ratios and four risk bands are research assumptions. These thresholds should be reestablished through cost estimates and inputs from relevant stakeholders.

D. Limitations

There are a few limitations to keep in mind. These include the non-immaculate programme-level test history, the lack of future rainfall and NWP inputs, the missing 2018 archive, and the incomplete 2026 record. Differences in station coverage and administrative danger levels may also affect data consistency. Neighbour/upstream variables are not entirely isolated across spatial folds, limiting spatial independence.

The comparison among deep learning models also has limitations: only the fusion configuration was re-run in a matched setting, and val_2022 was excluded during model selection. The strict development-event definition also differs from the lenient test-event definition, which may affect comparability. Finally, SHAP explanations show associations, not causal relationships.

With its admitted limitations, D16 is best viewed as a strong benchmark and a promising candidate for future evaluation. Further testing under operational conditions is necessary before it can be considered suitable for deployment as a flood-warning system.

VII. Conclusion

We present a leakage-aware probabilistic framework for 24/48/72 h FFWC danger-level risk prediction in Bangladesh. D16 is the strongest configuration on the 2022 validation set (mean PR-AUC: 0.9574) and achieves 0.7708 on the frozen test. It exceeds persistence results (PR-AUC: 0.6675) across all 21 test cell–horizon points. The top-performing DL model achieved a PR-AUC of 0.9356 on the validation set, and a matched fusion rematch remained 0.0206 below XGBoost. The model is well calibrated. Strict event recall improved over persistence at all horizons, particularly at 48–72 h, and TreeSHAP attributes 64% of 24 h importance to local hydrology. An intermediate in-sample scoring defect was

corrected during a protocol audit but did not change the selected model or the frozen-test result. Overall, reliable flood AI requires leakage-aware evaluation, strong hydrological baselines, and transparent test discipline. Future work should include forecast rainfall and potential operational evaluation.

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